

```

????????????????????????????????????????? */ :root, [data-theme="light"] { --bg-page:
#f8fafc; --bg-card: #ffffff; --bg-muted: #f1f5f9; --bg-header: #f8fafc;
--border: #e2e8f0; --text-1: #0f172a; --text-2: #475569; --text-3: #94a3b8;
--accent: #6366f1; --accent-bg: #eef2ff; --pass-text: #15803d; --pass-bg:
#f0fdf4; --warn-text: #b45309; --warn-bg: #fffbeb; --fail-text: #b91c1c; --fail-
bg: #fef2f2; --shadow: 0 1px 3px 0 rgb(0 0 0 / 0.07); } /* ?? Design Tokens:
Dark ?????????????????????????????????????????????????????????????????? */ [data-theme="dark"] {
--bg-page: #0f172a; --bg-card: #1e293b; --bg-muted: #1e293b; --bg-header:
#1e293b; --border: #334155; --text-1: #f1f5f9; --text-2: #94a3b8; --text-3:
#64748b; --accent: #818cf8; --accent-bg: #1e1b4b; --pass-text: #4ade80; --pass-
bg: #052e16; --warn-text: #fbbf24; --warn-bg: #1c1400; --fail-text: #f87171;
--fail-bg: #1c0505; --shadow: 0 1px 3px 0 rgb(0 0 0 / 0.3); } /* ?? Density
Scale ?????????????????????????????????????????????????????????????????? */ /* Default
(comfortable) spacing: these vars are overridden per density. */ :root { --d-py:
0.75rem; /* table cell vertical padding */ --d-px: 1rem; /* table cell
horizontal padding */ --d-gap: 1rem; /* grid/flex gap */ --d-card: 1.5rem; /*
card padding */ } [data-density="compact"] { --d-py: 0.35rem; --d-px: 0.625rem;
--d-gap: 0.5rem; --d-card: 1rem; font-size: 0.875rem; } [data-
density="spacious"] { --d-py: 1.125rem; --d-px: 1.375rem; --d-gap: 1.5rem;
--d-card: 2rem; } /* ?? Audience Gating
????????????????????????????????????????????????????????????????? */ /* Sections marked .aud-
manager are hidden for the executive audience. Sections marked .aud-engineer are
hidden for executive and manager. */ [data-audience="executive"] .aud-manager,
[data-audience="executive"] .aud-engineer { display: none; } [data-
audience="manager"] .aud-engineer { display: none; } /* ?? Base Styles
????????????????????????????????????????????????????????????????? */ *, *::before, *::after
{ box-
sizing: border-box; } body { background: var(--bg-page); color: var(--text-1);
font-family: ui-sans-serif, system-ui, -apple-system, sans-serif; margin: 0;
padding: 0; transition: background 0.2s ease, color 0.2s ease; } main { max-
width: 64rem; margin: 0 auto; padding: 2.5rem 1.5rem; } /* ?? Controls Bar
????????????????????????????????????????????????????????????????? */ .ctrl-bar { position:
sticky; top: 0; z-index: 20; background: var(--bg-card); border-bottom: 1px

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solid var(--border); padding: 0.5rem 1.5rem; display: flex; align-items: center;
gap: 0.75rem; flex-wrap: wrap; box-shadow: var(--shadow); } .ctrl-brand { font-
size: 0.7rem; font-weight: 700; letter-spacing: 0.15em; color: var(--accent);
margin-right: auto; text-transform: uppercase; } .ctrl-group { display: flex;
align-items: center; gap: 0.35rem; } .ctrl-label { font-size: 0.68rem; text-
transform: uppercase; letter-spacing: 0.08em; color: var(--text-3); } .ctrl-
select, .ctrl-btn { font-size: 0.78rem; padding: 0.2rem 0.5rem; border: 1px
solid var(--border); border-radius: 0.375rem; background: var(--bg-muted);
color: var(--text-1); cursor: pointer; transition: background 0.15s, border-
color 0.15s; } .ctrl-select:hover, .ctrl-btn:hover { background: var(--accent-
bg); border-color: var(--accent); color: var(--accent); } /* ?? Cards
???????????????????????????????????????????????????????????????? */ .card {
background: var(--bg-card); border: 1px solid var(--border); border-radius:
1rem; padding: var(--d-card); margin-bottom: 2rem; box-shadow: var(--shadow); }
.card-title { font-size: 1.125rem; font-weight: 600; color: var(--text-1);
margin: 0 0 1rem; } .card-subtitle { font-size: 0.95rem; font-weight: 600;
color: var(--text-1); margin: 1.25rem 0 0.625rem; } /* ?? KPI Grid
???????????????????????????????????????????????????????????????? */ .kpi-grid {
display: grid; gap: var(--d-gap); grid-template-columns: repeat(auto-fill,
minmax(9rem, 1fr)); margin-top: 1.25rem; } .kpi-cell { background: var(--bg-
muted); border-radius: 0.7
5rem; padding: var(--d-py) var(--d-px); } .kpi-label { font-size: 0.68rem; text-
transform: uppercase; letter-spacing: 0.1em; color: var(--text-3); margin: 0; }
.kpi-value { font-size: 1.1rem; font-weight: 600; color: var(--text-1); margin:
0.2rem 0 0; } /* ?? Tables
???????????????????????????????????????????????????????????????? */ .tbl-wrap {
border-radius: 0.75rem; overflow: hidden; border: 1px solid var(--border);
margin-top: 1rem; } table { width: 100%; border-collapse: collapse; background:
var(--bg-card); } thead tr { background: var(--bg-header); } thead th { padding:
var(--d-py) var(--d-px); text-align: left; font-size: 0.7rem; font-weight: 600;
text-transform: uppercase; letter-spacing: 0.06em; color: var(--text-2); } tbody
td { padding: var(--d-py) var(--d-px); color: var(--text-2); border-top: 1px
solid var(--border); font-size: 0.875rem; } /* Row-level status tints driven by
tr.pass / tr.warn / tr.fail classes. These interact with the CSS token system so
```

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they invert correctly in dark mode without any extra JavaScript. */ tbody
tr.pass td { background: var(--pass-bg); color: var(--pass-text); } tbody
tr.warn td { background: var(--warn-bg); color: var(--warn-text); } tbody
tr.fail td { background: var(--fail-bg); color: var(--fail-text); } /* ?? Status
Labels ?????????????????????????????????????????????????????????????????? */ .status-pass {
color: var(--pass-text); font-weight: 700; font-size: 0.7rem; text-transform:
uppercase; letter-spacing: 0.08em; } .status-warn { color: var(--warn-text);
font-weight: 700; font-size: 0.7rem; text-transform: uppercase; letter-spacing:
0.08em; } .status-fail { color: var(--fail-text); font-weight: 700; font-size:
0.7rem; text-transform: uppercase; letter-spacing: 0.08em; } /* ?? Badge
????????????????????????????????????????????????????????????????????????? */ .badge {
display: inline-block; border-radius: 999px; padding: 0.15rem 0.625rem; font-
size: 0.68rem; font-weight: 600; letter-spacing: 0.06em; } .badge-accent {
background: var(--accent-bg);
color: var(--accent); } /* ?? Inline code
????????????????????????????????????????????????????????????????????????? */ code { background:
var(--bg-muted); border-radius: 0.25rem; padding: 0.1em 0.35em; font-size:
0.85em; color: var(--accent); } BLSN Theme ? Light ? Dark ? Auto Density Compact
Comfortable Spacious Audience Executive Manager Engineer BLSN Synthetic
Validation Report Theme&nbsp; light &nbsp;·&nbsp;Density&nbsp; comfortable
&nbsp;·&nbsp;Audience&nbsp; engineer Vehicle P3 Orion Cooling Mode
helium_closed_loop Nominal Flow 110 g/s Limits 80?140 g/s Sanity Check Telemetry
Status Message PASS [PASS] PASS | Nominal mass flow is bounded by configured
limits System Verification: Claimed vs. Actual Claimed Limits: 0 &nbsp;|&nbsp;
Actually Tested: 0 &nbsp;? STATUS: UNKNOWN Total Runtime 0.00s Pass Ratio 0.0%
Fail Ratio 0.0% Skip Ratio 0.0% Tuple Execution Telemetry Targets (seconds)
Metric Target Warn Fail ci_total_runtime_s 120.0 180.0 240.0 pytest_runtime_s
15.0 25.0 35.0 pipeline_runtime_s 5.0 10.0 15.0 yaml_json_lint_runtime_s 3.0 6.0
9.0 Interactive Visualizations Telemetry trend and PCA bottleneck analysis ?
theme-aware rendering. System Priority & Bottleneck Ranking (Bradley-Terry)
Pairwise probability model:  $P(i \text{ \> } j) = \frac{p_i}{(p_i + p_j)}$  Global Repo Ranking
Rank Artifact BT Strength Win Prob vs Top 1 docs/index.html 2.533414 50.00% 2
docs/_includes/metric_card.html 2.317756 47.78% 3 docs/presentation.html
2.309540 47.69% 4 docs/_includes/plotly_render.html 2.246689 47.00% 5
```

docs/\_layouts/default.html 2.161844 46.04% 6 docs/blsn\_report.pdf 0.795024  
23.89% 7 docs/visuals/plotly\_payloads.json 0.481794 15.98% 8  
docs/experimental\_design\_matrix.json 0.417307 14.14% 9  
docs/historical\_telemetry\_ledger.json 0.409257 13.91% 10  
docs/federation/federation\_cherry\_pick.json 0.382788 13.13% Inter-Category  
Ranking Rank Category Win Prob vs Top 1 html 50.00% 2 json 15.19% 3 md 10.52% 4  
pdf 6.43% 5 yaml 3.18% 6 yml 2.87% QPLANT Requirements Ranking Bridge Rank  
Requirement Win Prob vs  
Top 1 ci\_total\_runtime\_s 50.00% 2 pytest\_runtime\_s 50.00% 3 pipeline\_runtime\_s  
50.00% 4 yaml\_json\_lint\_runtime\_s 50.00% Wave 12: SPC & Telemetry Stats  
Runtime Drift 0.0000s Prev Moving Avg 0.0000s Total Runs 2 Success Rate 100.0%  
DoE Factors ? A: python\_version · B: os\_type · C: load\_parameter Principal  
Component Analysis (PCA Proof) Principal Component Eigenvalue Explained Variance  
Highest Weight Variable PC1 0.000000 0.00% total\_tests PC2 0.000000 0.00%  
skip\_rate Metric Markdown Slugs Ci Total Runtime S Pytest Runtime S Pipeline  
Runtime S Yaml Json Lint Runtime S Ci Total Runtime S Metric ID:  
`ci\_total\_runtime\_s` Target Threshold: \*\*120.0s\*\* Warn Threshold: \*\*180.0s\*\*  
Fail Threshold: \*\*240.0s\*\* Pytest Runtime S Metric ID: `pytest\_runtime\_s` Target  
Threshold: \*\*15.0s\*\* Warn Threshold: \*\*25.0s\*\* Fail Threshold: \*\*35.0s\*\*  
Pipeline Runtime S Metric ID: `pipeline\_runtime\_s` Target Threshold: \*\*5.0s\*\*  
Warn Threshold: \*\*10.0s\*\* Fail Threshold: \*\*15.0s\*\* Yaml Json Lint Runtime S  
Metric ID: `yaml\_json\_lint\_runtime\_s` Target Threshold: \*\*3.0s\*\* Warn Threshold:  
\*\*6.0s\*\* Fail Threshold: \*\*9.0s\*\* Mermaid System Flow Diagram graph LR;  
SSOT[SSOT Config] --> PIPELINE[Pipeline Execution]; PIPELINE -->  
MARKDOWN[Markdown Slug Pages]; PIPELINE --> INDEX[Index HTML]; PIPELINE -->  
SLIDES[Slides HTML]; INDEX --> PDF[PDF Export]; /\* ?? BLSN Report UI Controls  
???????????????????????????????????????? All user preferences are persisted to  
localStorage under the "blsn\_" namespace so they survive hard reloads without  
server round-trips. Theme: light | dark | auto "auto" resolves to light or dark  
via prefers-color-scheme and keeps the attribute updated when the OS preference  
changes (e.g. switching between light and dark system themes). Density: compact  
| comfortable | spacious Controls the --d-py / --d-px / --d-gap / --d-card CSS  
tokens that scale table cell padding and card gutters without reflowing the  
page. Audience: executive | manager | engineer CSS [data-audience] selectors

hide

.aud-manager and .aud-engineer sections for less technical readers. Changing this value simply updates the data attribute; no DOM nodes are added or removed.

Plotly charts are re-rendered via `Plotly.react()` rather than `Plotly.newPlot()`

when the theme changes so the existing DOM element is reused and transitions

```

feel instant. ?? */ (function () { 'use strict'; const PREF_NS = 'blsn_'; const

```

```
getPref = (k, def) => localStorage.getItem(PREF_NS + k) || def; const setPref =
```

```
(k, v) => localStorage.setItem(PREF_NS + k, v); const root =
```

```
document.documentElement; /* Resolve the "auto" theme sentinel using the OS
```

```

preference. */ function resolveTheme(row) { if (row !== 'auto') return row;

```

```
return window.matchMedia('(prefers-color-scheme: dark)').matches ? 'dark' :
```

```
'light'; } /* Apply a data-attribute and keep its matching in sync. */ function
```

```
applyAttr(attr, selectId, value) { root.setAttribute('data-' + attr, value);
```

```
const el = document.getElementById(selectId); if (el) el.value = value;
```

```
setPref(attr, value); } /* ?? Plotly theme-aware rendering
```

```
???????????????????????????????????????? */ let  payload = null; /* cached payload to
```

allow re-render on theme change \*/ /\* Map report theme to the matching built-in

```
Plotly template name. */ const plotlyTpl = (themeRaw) => resolveTheme(themeRaw)
```

```
=== 'dark' ? 'plotly_dark' : 'plotly_white'; /* Patch a Plotly chart layout to
```

```
inherit the current theme colours. */ function patchLayout(chart, themeRaw) {
```

```
const isDark = resolveTheme(themeRaw) === 'dark'; return Object.assign({},
```

```
chart.layout || {}, { template: plotlyTpl(themeRaw), paper_bgcolor:
```

```
'rgba(0,0,0,0)', plot bgcolor: 'rgba(0,0,0,0)', font: { color: isDark ?
```

```
'#94a3b8' : '#475569' }, xaxis: Object.assign({}, (chart.layout || {}).xaxis, {
```

```
gridcolor: isDark ? '#334155' : '#e2e8f0' }, yaxis: Object.assign({},
```

```
(chart.layout || {}).yaxis, { gridcolor: isDark ? '#334155' : '#e2e8f0' }, );
```

```
}/* Re-render both Plotly charts with the resolved theme.*/ function
```

```
rerenderPlotly(themeRaw) { if (!window
```

```
.Plotly || ! payload) return; const charts = payload.charts || {}; const tel =
```

```
charts.telemetry combo || {}; const pca = charts.pca bottlenecks || {};
```

```
Plotly.react('plotly-telemetry-chart', tel.data || [], patchLayout(tel,
```

```
themeRaw)); Plotly.react('plotly-pca-chart', pca.data || [], patchLayout(pca,
```

```
themeRaw)); } /* ?? Public setters wired to onchange handlers
```

```

???????????????????? */ window.setTheme = function (v) { /* Apply the resolved

```

```

value to data-theme but store the raw choice (including "auto") so the select
always reflects user intent. */ root.setAttribute('data-theme',
resolveTheme(v)); const el = document.getElementById('ctrl-theme'); if (el)
el.value = v; setPref('theme', v); rerenderPlotly(v); }; window.setDensity = (v)
=> applyAttr('density', 'ctrl-density', v); window.setAudience = (v) =>
applyAttr('audience', 'ctrl-audience', v); /* ?? Initialise from saved
preferences (fall back to SSOT defaults) */ const initTheme = getPref('theme',
'light'); const initDensity = getPref('density', 'comfortable'); const
initAudience = getPref('audience', 'engineer'); /* Apply theme without going
through the setter so the select shows "auto" rather than the resolved value
when stored as "auto". */ root.setAttribute('data-theme',
resolveTheme(initTheme)); const themeEl = document.getElementById('ctrl-theme');
if (themeEl) themeEl.value = initTheme; applyAttr('density', 'ctrl-density',
initDensity); applyAttr('audience', 'ctrl-audience', initAudience); /* ?? Plotly
payload fetch ?????????????????????????????????????????????????????????????? */ const emptyLayout =
(themeRaw) => ({ template: plotlyTpl(themeRaw), paper_bgcolor: 'rgba(0,0,0,0)',
plot_bgcolor: 'rgba(0,0,0,0)', xaxis: { visible: false }, yaxis: { visible:
false }, annotations: [{ text: 'No Plotly payload available', showarrow: false,
xref: 'paper', yref: 'paper', x: 0.5, y: 0.5, }], });
fetch('visuals/plotly_payloads.json') .then((r) => (r.ok ? r.json() :
Promise.reject(new Error('HTTP ' + r.status)))) .then((payload) => { _payload
= payload; rerenderPlotly(getPref('theme', 'light')); }) .catch(() => { const
el = emptyLayout(getPref('theme', 'light')); Plotly.newPlot('plotly-telemetry-
chart', [], el); Plotly.newPlot('plotly-pca-chart', [], el); }); /* Keep "auto"
theme in sync when the OS preference changes at runtime (e.g. user switches
system dark mode while the tab is open). */ window.matchMedia('(prefers-color-
scheme: dark)').addEventListener('change', () => { const stored =
getPref('theme', 'light'); if (stored === 'auto') { root.setAttribute('data-
theme', resolveTheme('auto')); rerenderPlotly('auto'); } }); });

```